Cal. Code Regs. Tit. 14, § 1724.7 - Project Data Requirements

(a) An underground injection project shall be supported by data filed with the Division that demonstrates to the Division's satisfaction that injected fluid will be confined to the approved injection zone and that the underground injection project will not cause damage to life, health, property, or natural resources. The engineering study, geologic study, and injection plan described in subdivisions (a)(1) through (a)(3) shall demonstrate to the Division's satisfaction that injected fluid will not migrate out of the approved injection zone through another well, geologic structure, fault, fracture, fissure, hole in the casing, or other pathway, considering project duration, volume of fluid to be injected, and other relevant factors. The operator is responsible for ensuring that the data are current and accurately reflect the project setting and operation throughout the operating life of the project. The data filed with the Division shall include, at a minimum:

(1) An engineering study, including but not limited to:

(A) A description of how the area of review was determined, including calculations, variables, citations, and assumptions.

(B) A map of the area of review showing the location of the following:

(i) All wells within and adjacent to the boundary of the area of review;

(ii) All water supply wells that are within the area of review and identified in public records or otherwise known to the operator;

(iii) Any underground disposal horizons, mining, and other subsurface industrial activities not associated with oil and gas production within the area of review, to the extent such information is publicly available or otherwise known to the operator; and

(iv) Traces of the geologic cross sections provided under subdivision (a)(2)(E).

(C) A compendium of the following information:

(i) For all wells depicted in subdivision (a)(1)(B) (including water supply wells to the extent information is known or publicly available), the API numbers, or other identifying information for wells that do not have API numbers, and the wellbore paths, total depths, and depths of completion interval(s) of the wells;

(ii) The type and status of water supply wells depicted in subdivision (a)(1)(B); and

(iii) All data specified in Section 1724.7.1, provided in the form of graphical casing diagrams or flat-file data sets, for all wells that are within the area of review and that are completed in or penetrating the injection zone for the underground injection project or a deeper zone, including directionally drilled wells that intersect the area of review in the injection zone or a deeper zone.

(D) The planned well-drilling and plugging-and-abandonment program to complete the project, including a flood-pattern map, if applicable, showing all injection, production, and plugged and abandoned wells, and unit boundaries.

(2) A geologic study, including but not limited to:

(A) Reservoir characteristics of the injection zone, such as porosity, permeability, average thickness, areal extent, fracture gradient, original and present temperature and pressure, and original and residual oil, gas, and water saturations. The scope of the geologic characterization shall

encompass the caprock and sealing mechanisms, the injection zone including the vertical interval above and below the approved injection zone, and the areas where potential migration of fluid or entrapment of migrated fluid could occur.

(B) Reservoir fluid data for the injection zone, such as oil gravity and viscosity, water quality, presence and concentration of non-hydrocarbon components in the associated gas (such as hydrogen sulfide), and specific gravity of gas. Liquid analysis of the reservoir fluid shall be performed in accordance with Section 1724.7.2.

(C) Structural contour map drawn on a geologic marker at or near the top and base of each injection zone in the area of review, indicating faults and any lateral containment features. If faults are identified, the operator must address whether or not the faults are capable of confining fluid to the approved injection zone, and any geologic features that could result in the migration of fluid out of the approved injection zone.

(D) Isopach map of each injection zone or subzone in the area of review.

(E) At least two geologic cross sections in the area of review through at least three wells, including one injection well. As near as possible, one of the geologic cross sections shall be along strike and the other geologic cross section shall be perpendicular to strike. The cross sections shall extend from the base of the deepest production or injection zone to surface and indicate the location of the approved injection zone, the base of freshwater, and the base of the USDW.

(F) Representative electric log to a depth below the deepest producing or injection zone, whichever is deeper, identifying all geologic units, formations, USDWs, freshwater aquifers, and oil or gas zones. The electric log shall identify the API number of the well that was logged.

(3) An injection plan, including but not limited to:

(A) Statement of primary purpose of the project.

(B) A map showing injection facilities related to the project, and piping and instrumentation diagram(s) for the injection facilities.

(C) Statement of the anticipated project duration, anticipated daily rate of injection (by well), and anticipated cumulative net volume of fluid to be injected.

(D) Identification of all wells that are part of the underground injection project, including injection wells, affected production wells, water source

wells, observation or other wells, and any planned wells to the extent known. The depths of water source wells shall also be provided.

(E) Monitoring system, including methods or standard operating procedures to be utilized to ensure that no damage is occurring and that the injection fluid is confined to the approved injection zone. In the event the Division, the State Water Resources Control Board, or the Regional Water Quality Control Board requires groundwater monitoring in relation to the underground injection project, or as a condition of project approval, the operator shall consult with the State Water Resources Control Board or the Regional Water Quality Control Board and provide the Division with documentation and the results of such consultation.

(F) Method of injection, including such information as injection string configuration and bottom-hole assembly.

(G) List of cathodic protection or other corrosion prevention measures employed for plant, lines, and wells, if such measures are warranted.

(H) Identification of the source(s) of the injection liquid and an analysis of the injection liquid, in accordance with Section 1724.7.2.

(4) All data supporting the determination under Section 1724.10.3 of the maximum allowable surface injection pressure for each injection well in the underground injection project, including all calculations, variables, citations, and assumptions.

(5) Copies of letters of notification sent to offset operators.

(6) Any other data that, in the judgment of the Division, are pertinent and necessary for the proper evaluation of the underground injection project. Examples of such data are: isogor maps, water-oil ratio maps, isobar maps, three-dimensional geologic models, reservoir simulation results, isopach maps of the confining layers, equipment diagrams, and safety programs.

(b) The addition of an injection well to an underground injection project is subject to approval by the Division, and shall be indicated in a summary list of approved injection wells associated with the underground injection project, which shall be referenced by the Project Approval Letter for the underground injection project. When an injection well is added to an underground injection project, the operator shall provide the Division with a brief description of how the injection well will impact the underground injection project, any data relevant to the addition of the injection well, and an update to the data previously provided to the Division if relevant conditions have changed or

if more accurate data have become available. The addition of an injection well does not require the operator to submit data previously provided to the Division.

(c) All data required under this section shall be submitted to the Division digitally. If requested by the Division, a hard copy of specified data shall also be submitted. All maps, diagrams, and exhibits required in subdivision (a) shall be clearly and appropriately labeled, such as to title, scale, and purpose, and shall clearly identify wells, boundaries, zones, contacts, and other relevant data.

(d) All data required under this section shall be submitted to the Division with a cover page including a statement that appropriate licensed professionals, whose signatures and stamps appear at the bottom of the page, are responsible for all data, if any, subject to the requirements of Business and Professions Code sections 6735, 7835, or 7835.1. If the operator determines that the submission does not include material subject to the requirements of Business and Professions Code sections 6735, 7835, or 7835.1, the cover page shall so indicate, and shall provide the name(s) and signature(s) of the individual(s) responsible for preparing the submission.

(e) The Division may accept data alternative to what is required under subdivision (a), provided that the operator demonstrates to the Division's satisfaction all of the following:

(1) It would be an unreasonable burden to provide the data specified in subdivision (a);

(2) The alternative data provided by the operator accomplishes the same purpose as the data it would replace;

(3) The underground injection project is, on whole, supported by data demonstrating that injected fluid will be confined to the approved injection zone, and that the underground injection project conforms to the requirements of this subchapter and will not cause damage to life, health, property, or natural resources.

Notes

Cal. Code Regs. Tit. 14, § 1724.7
1. New section filed 2-17-78; effective thirtieth day thereafter (Register 78, No. 7).
2. Amendment filed 12-28-84; effective thirtieth day thereafter (Register 84, No. 52).
3. Change without regulatory effect amending subsections (a)(4)-(5) filed 9-3-96 pursuant to section 100, title 1, California Code of Regulations (Register 96, No. 36).
4. Amendment filed 2-6-2019; operative 4-1-2019 (Register 2019, No. 6).

Note: Authority cited: Sections 3013 and 3106, Public Resources Code. Reference: Section 3106, Public Resources Code.

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